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Chapter 17

GELFAND-P000900

[source number: 3]

21.1.80. Gelfand Sem.

A. Migdal. On a new approach to gauge theories.

This is a new language, more adequate for describing strong interactions.

Old language: $SU(N)$

fields

$$\underset{\text{(connection)}}{A_\mu(x)} = \sum_{a=1}^{N^2-1} T_a A_\mu^a, \qquad T_a \text{ are the generators of } SU(N). \tag{1}$$

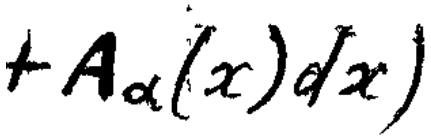
The language would be adequate if the fields were close to 0 or to another classical solution (if they did not fluctuate so strongly).

We need to consider.

$$\Psi(c) = \operatorname{Tr} p \exp\left(\oint A\,dx\right), \tag{2}$$

where c is a loop,

$$p \exp\left(\oint A\,dx\right) = \prod (1 + A_\alpha(x)\,dx) \tag{3}$$



Uncertain reading. Equation (3) reads $A_\alpha(x)\,dx$, with no visible index on dx.

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Without the trace in (2), a [crossed out] covariant quantity.



Uncertain reading. The crossed-out entry before «ковар.» (covar.) may begin «ков.».

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It turns out that we can restrict ourselves to what results when we take the trace.
Thus we need loop space.
Noninvariant quantities come out equal to 0 in quantum theory.

— • —

I.M.G. The point is that a repr. of the group of paths is determined by its character.

— · —

Here we are speaking of the lowest character.
Parallel transport:

$$\frac{dF}{dx} = AF \quad \Rightarrow \quad F = Ce^{\int A\,dx}$$

for matrices the integral is ordered.
Group of paths.

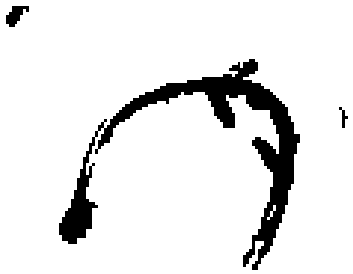


Diagram and typed description. An arc with two arrowheads. The strokes on the right branch are close together.



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we cancel the paths going out and back.

— · —

Let A_μ satisfy the classical equations of motion:

$$\nabla_\mu F_{\mu\nu} = 0, \tag{4}$$

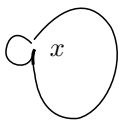
where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \tag{5}$$

(curvature of the connection)
Let us write the condition satisfied in this case by the function $\Psi(c)$.



Diagram and typed description. A small loop at x, with a thickened stroke at the join.

(6) 

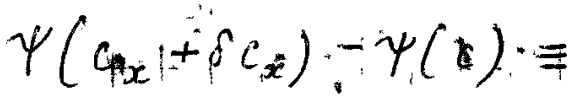
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$c + \delta\sigma_{\mu\nu}$, $\delta\sigma_{\mu\nu}$ is the area (we have attached a small loop to the contour c)

$$\frac{\delta\Psi}{\delta\sigma_{\mu\nu}} = \mathrm{Tr}\left(\hat{p}F_{\mu\nu}\exp\left(\oint A\,dx\right)\right) \tag{7}$$

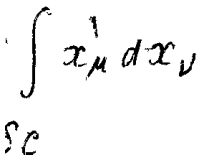
Let us prove (7). We have

$$\Psi(c_x + \delta c_x) - \Psi(c) \equiv \frac{\delta\Psi}{\delta\sigma_{\mu\nu}}\int_{\delta c} x'_\mu\,dx_\nu \tag{8}$$



Uncertain reading. The first c_x is heavily overwritten. The argument of the second Ψ resembles a corrected c or δ.

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Uncertain reading. The detached stroke above x_μ is rendered as a prime on x.

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Stokes-type functional. if the variation is proportional to the area (this is not always so, e.g. for length)

GELFAND-P000901



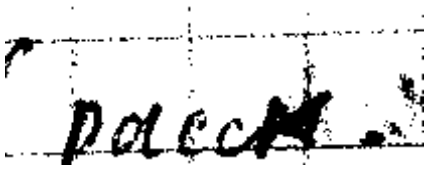
Uncertain reading. Unclear mark in the upper-left margin, resembling a question mark.

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In the Abelian case this is Stokes's theorem.

$$\oint_{\delta C} A_\alpha\,dx_\alpha = \int_{\delta S} F_{\alpha\beta}\,d\sigma_{\alpha\beta}. \tag{9}$$

(discusses work with Makeenko and uses work by the authors Mandelstam, Nambu, Polyakov)



Uncertain reading. The first abbreviation may mean 'discusses' or 'considers'.

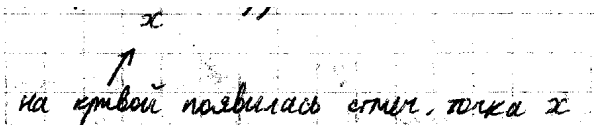
Original source details: GELFAND-P000901

$\hat{p}$ in (7) acts on both $F_{\mu\nu}$ and exp.

$F_{\mu\nu}(x)$ is placed where the point x is

Or:

$$\frac{\delta\Psi}{\delta\sigma_{\mu\nu}} = \mathrm{Tr}\left(F_{\mu\nu}(x)\, p \exp\left(\int_x^x A\, dx\right)\right) \tag{10}$$



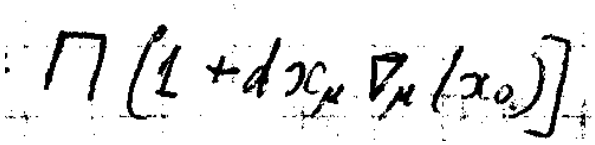
Original source details: GELFAND-P000901

Diagram and typed description. Annotation with an arrow beneath the integral in (10). The abbreviation before 'point' is read tentatively as 'marked'.

↗ a marked point x has appeared on the curve

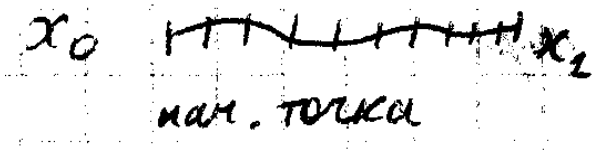
Proof of (7):

$$\begin{aligned}\prod(1 + A(x)\, dx) &= \prod\left(1 + dx_\mu\left(A_\mu(x_0) + \frac{\partial}{\partial x_\mu}\right)\right) \\ &= \prod(1 + dx_\mu \nabla_\mu(x_0)]\end{aligned} \tag{11}$$



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Uncertain reading. The last factor in (11) opens with a parenthesis and ends with a square bracket.

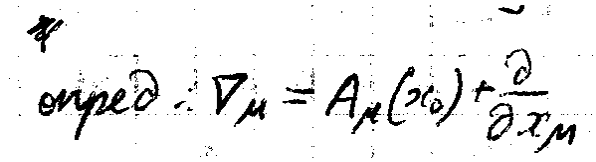


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Diagram and typed description. Subdivision of the curve beneath (11).

x_0 ⊢⊣ x_1

initial point

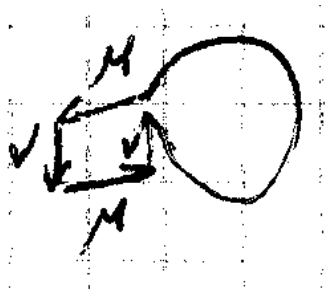


Original source details: GELFAND-P000901

Diagram and typed description. Definition of ∇_μ with an arrow toward (11).

↗ def. $\nabla_\mu = A_\mu(x_0) + \frac{\partial}{\partial x_\mu}$

$dx_\mu \frac{\partial}{\partial x_\mu}$ shifts the argument



Original source details: GELFAND-P000901

Diagram and typed description. Loop with the rectangular circuit (12).

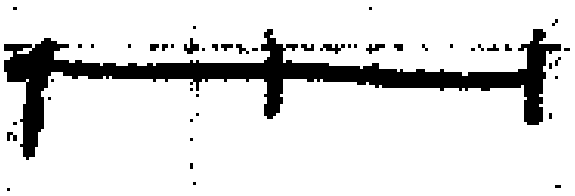
(12) μ, ν, ν, μ

— consider the following variation

$$(1 + dx_\mu \nabla_\mu)(1 + dx_\nu \nabla_\nu)(1 - dx_\mu \nabla_\mu)(1 - dx_\nu \nabla_\nu) = \\ = 1 + dx_\mu \wedge dx_\nu [\nabla_\mu, \nabla_\nu] \tag{13}$$

(here all ∇_μ refer to the initial point)

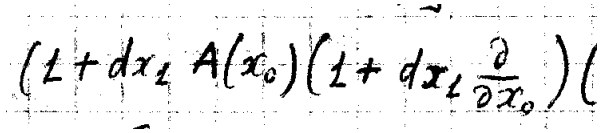
Take two segments:



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Diagram and typed description. Two consecutive segments.

$$(1 + A(x_0)\, dx_1)(1 + A(x_0 + dx_1)\, dx_2) = \\ = \left[1 + dx_1 \left(A(x_0) + \frac{\partial}{\partial x_0}\right)\right](1 + A(x_0)\, dx_2) = \\ = \left(1 + dx_1 A(x_0)\left(1 + dx_1 \frac{\partial}{\partial x_0}\right)(\dots)\right. \tag{14}$$



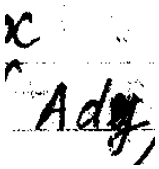
Original source details: GELFAND-P000901

Uncertain reading. No separate parenthesis closing the first factor is visible after $A(x_0)$ in the last line of (14).

(∼ transport around a closed contour is curvature)

Differentiate (10) with respect to x (without thinking, for now):

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\left(\{\partial_\mu F_{\mu\nu}(x) + [A_\mu(x), F_{\mu\nu}]\} \right. \\ \left. p \exp \int_x^x A\, dy\right) \tag{15}$$



Original source details: GELFAND-P000901

Uncertain reading. The differential after A in (15) is overwritten. dy is the preferred reading, with dx also possible.

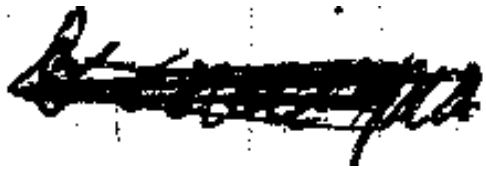
commutator due to cyclic invariance of the trace

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[source number: 4]

Geometrically:

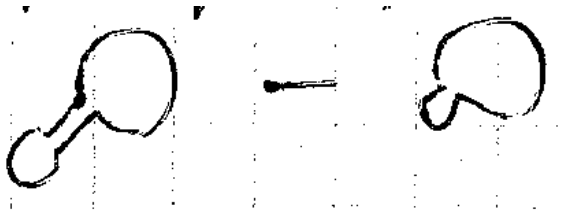
we shift the attached loop



Original source details: GELFAND-P000902

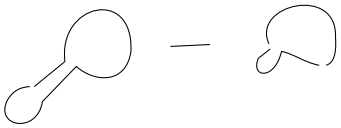
Uncertain reading. The word before ‘by an amount’ is crossed out. Individual strokes are visible and the reading is unresolved.

by an amount dx.



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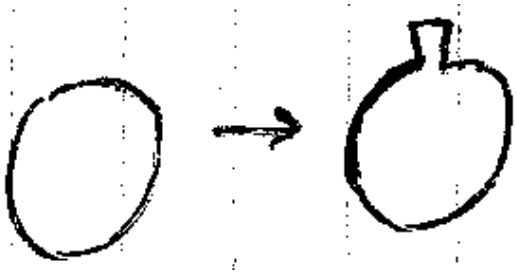
Diagram and typed description. Two positions of the attached loop.



I.M.G. A repr. of the group of paths is called a connection.

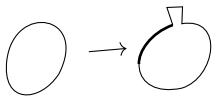
Lie alg.?

Curvature — something like a Lie algebra.

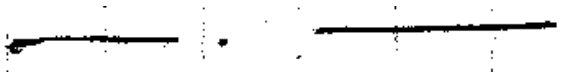


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Diagram and typed description. An arrow from a loop to a loop with a protrusion.



— like a shift by an element of the Lie algebra.

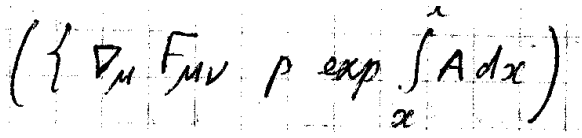


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Diagram and typed description. Manuscript divider.

A.M. The braces $\{\ldots\}$ (in (15)) vanish by the classical eqs. of motion.

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\left(\{ \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dx \right) \tag{15'}$$



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Uncertain reading. In (15′) a brace opens before the derivative. The expression ends with a parenthesis.

In fact

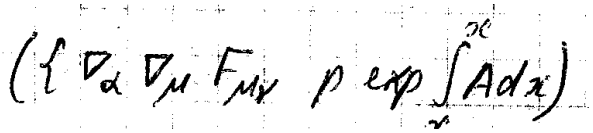


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Uncertain reading. The mark before ‘one can’ is crossed out. Part of its upper arc is visible and the reading is unresolved.

one can further add:

$$\partial_\alpha \partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\left(\{ \nabla_\alpha \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dx \right) \tag{16}$$



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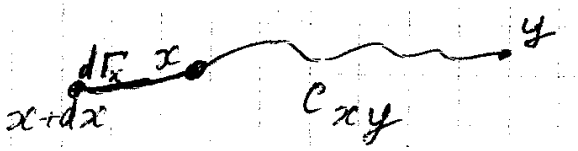
Uncertain reading. In (16) a brace opens before the derivatives. The expression ends with a parenthesis.

The left-hand side is noncommutative!! (since it is a functional of the point and also of the path)

$$\frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \hat{\varphi}_{\mu\nu}(C_x) \tag{17}$$

— a function on contours passing through the point x

The left-hand side of (15) needs a definition:



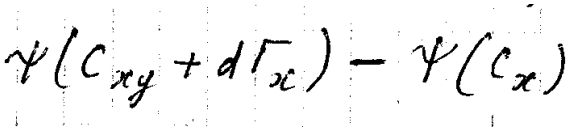
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Diagram and typed description. The contour C_{xy} and the short segment $d\Gamma_x$.



$d\Gamma_x$ — the contour going from x to $x + dx$.

$$\Psi(C_{xy} + d\Gamma_x) - \Psi(C_x) = dx_\mu \frac{\partial \Psi}{\partial x_\mu} \tag{18}$$



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Uncertain reading. The first argument contains C_{xy}, the second C_x.

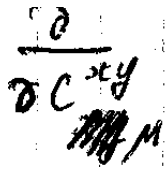
— this defines the right-hand side (and hence $\frac{\partial \Psi}{\partial x_\mu}$).

The contour passes through the point x twice: we shift both the beginning and the end

For C_{xx}, the beginning and the end are shifted in turn.

Translate the contour:

$$\frac{\partial}{\partial C_\mu^{xy}} = \frac{\partial}{\partial x_\mu} + \frac{\partial}{\partial y_\mu} + \int_x^y dz_\alpha \frac{\delta}{\delta \sigma_{\alpha\mu}(z)} \tag{19}$$



Uncertain reading. An index is crossed out before μ in the left denominator of (19).

Original source details: GELFAND-P000902

(translation of the contour)

For transl. invariant functionals, we get:

$$\partial_\mu = \oint dz_\nu \frac{\delta}{\delta \sigma_{\mu\nu}(z)}. \tag{20}$$

(for closed contours)

GELFAND-P000903

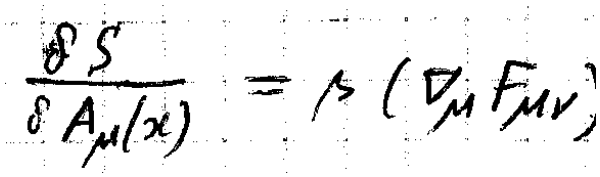
In the classical case we obtain

$$\mathcal{L} = \beta \operatorname{Tr} F_{\mu\nu}^2 \tag{21}$$

$$S = \beta \int \operatorname{Tr} F_{\mu\nu}^2 \, d^4x \tag{22}$$

As is known, we have:

$$\frac{\delta S}{\delta A_\mu(x)} = \beta(\nabla_\mu F_{\mu\nu}) \tag{23}$$



Uncertain reading. The denominator at left reads $A_\mu(x)$ and the right-hand side has $\nabla_\mu F_{\mu\nu}$.

Original source details: GELFAND-P000903

Therefore (15) $\Rightarrow$

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = 0 \tag{24}$$

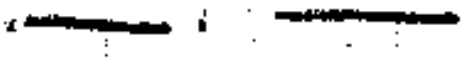
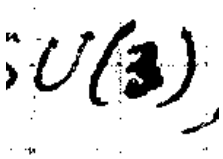


Diagram and typed description. Manuscript divider.

Original source details: GELFAND-P000903

The $SU(\infty)$ theory is close to an exact solution

(expansion in powers of $\frac{1}{n}$; in reality $SU(3)$)

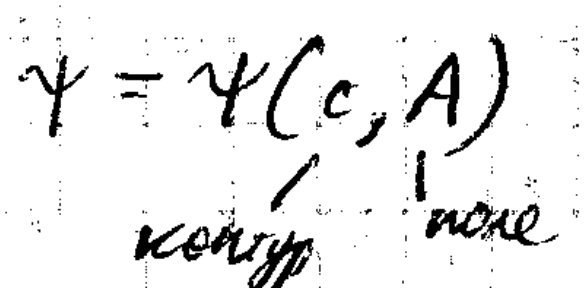


Original source details: GELFAND-P000903

Uncertain reading. The digit in $SU(3)$ is heavily traced over. It reads as 3.

Let us derive the quantum eq. of motion.

QFT Consider



Original source details: GELFAND-P000903

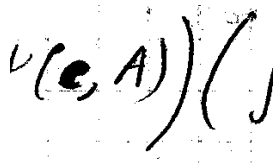
Diagram and typed description. Labels on the two arguments.

$\Psi = \Psi($ c, $A)$

contour field

and now consider

$$\Psi(c) = \left(\int \mathcal{D}A_\mu \, \exp(-S)\, \Psi(c, A)\right)\left(\int \mathcal{D}A_\mu \, \exp(-S)\right) \tag{25}$$



Original source details: GELFAND-P000903

Uncertain reading. In (25) the two parenthesized groups are written next to each other. No separate division sign is visible between them. The c in $\Psi(c, A)$ is heavily traced over.

Quantum eq. of motion: a property of the integral

$$\int \mathcal{D}A_\mu(x) \frac{\delta}{\delta A_\mu(x)} \{e^{-S} Q[A]\} \equiv 0 \tag{26}$$

$Q[A]$ — an arbitrary functional.

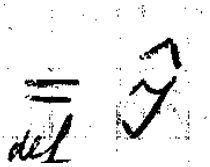
From (26) it follows that

$$\left\langle \frac{\delta S}{\delta A_\mu(x)} Q \right\rangle = \left\langle \frac{\delta Q}{\delta A_\mu(x)} \right\rangle \tag{27}$$

(Schwinger eq.)

From this, omitting Q and introducing $\overline{h}$, we obtain:

$$\frac{\delta S}{\delta A_\mu(x)} = \overline{h} \frac{\delta}{\delta A_\mu(x)} \underset{\text{def}}{=} \hat{\mathcal{J}} \tag{28}$$

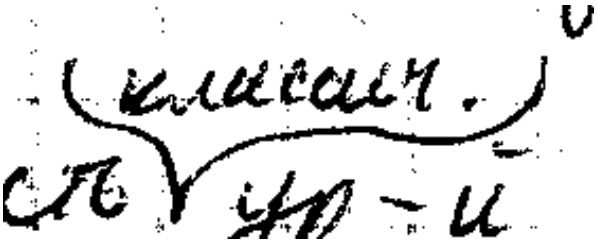


Original source details: GELFAND-P000903

Uncertain reading. The handwritten sign under the hat is read tentatively as $\mathcal{J}$.

Or:

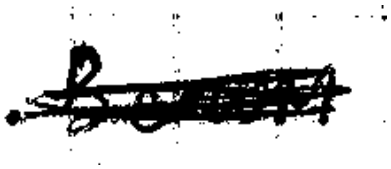
the left-hand side of the [inserted above: classical] eqs. of motion is replaced by variation with respect to the field.



Original source details: GELFAND-P000903

Diagram and typed description. The insertion 'classical' above 'eqs.'

$$\underbrace{\text{eqs.}}_{\text{classical}}$$



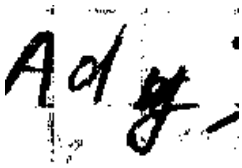
Original source details: GELFAND-P000903

Uncertain reading. The crossed-out word may begin with the Cyrillic letter «В». The remainder is unresolved.

The fields Ψ are not independent (much redundant information).

$(15') \Rightarrow$

$$\begin{aligned} \beta\, \partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}} &= \beta \left\langle \operatorname{Tr} \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dy \right\rangle = \\ &= \left\langle \operatorname{Tr} \frac{\delta}{\delta A_\mu(x)}\, p \exp\left(\int_x^x A\, dy \right) \right\rangle, \qquad \beta = \overline{h}^{-1}. \end{aligned} \tag{29}$$



Original source details: GELFAND-P000903

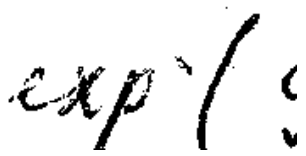
Uncertain reading. In the first integral in (29), the last letter in dy is overwritten. The second integral clearly reads dy.

GELFAND-P000904

[source number: 5]

Here

$$\Psi(c, A) = \operatorname{Tr} p \exp\left(\oint_c A\, dx\right) \tag{30}$$



Original source details: GELFAND-P000904

Uncertain reading. A separate short stroke of uncertain significance is visible between exp and the opening parenthesis.

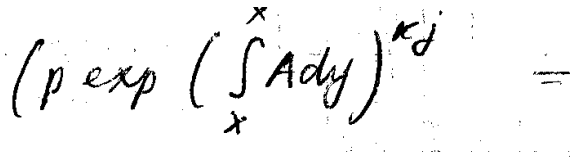
[Crossed out: $\oint_c dy$.]



Original source details: GELFAND-P000904

Uncertain reading. The crossed-out entry reads as a contour integral with a lower label and a differential.

$$\frac{\delta}{\delta A_\mu^{jk}(x)}\left(p \exp\left(\int_x^x A\, dy\right)^{kj} = \right.$$
$$= \oint_c dy_\nu\, U^{ii}(c_{xy}) U^{kk}(c_{yx}) \delta^4(x-y), \tag{31}$$



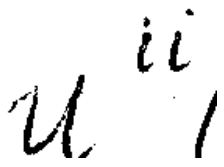
Original source details: GELFAND-P000904

Uncertain reading. In the first line of (31), the outer opening parenthesis is unpaired. An equals sign follows the superscripts.



Original source details: GELFAND-P000904

Uncertain reading. A short entry is crossed out above the integral sign. Its reading is unresolved.



Original source details: GELFAND-P000904

Uncertain reading. The first matrix U on the right of (31) has two superscript i's, each with a separate dot.

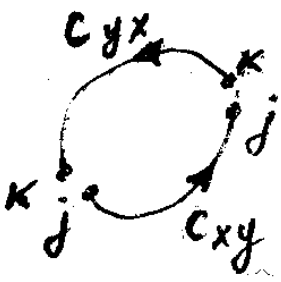
(32) where U is the transport matrix



Original source details: GELFAND-P000904

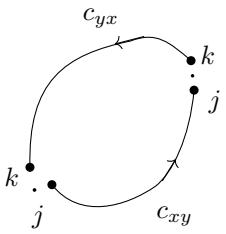
Uncertain reading. A short crossed-out mark appears between 'transport' and the formula for U.

$$U(C_{xy}) = P \exp \int_x^y A\, dx \qquad \delta^4(x-y) = \frac{\delta A(y)}{\delta A(x)}$$



Original source details: GELFAND-P000904

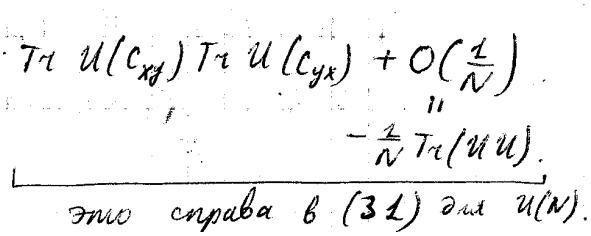
Diagram and typed description. Arcs c_{yx} and c_{xy} with their directions and the labels j, k.



For an arbitrary (semisimple) group:

$$\sum_a \operatorname{Tr}\big(T^a U(c_{xy}) T^a U(c_{yx})\big) = \operatorname{Tr} U(c_{xy}) \operatorname{Tr} U(c_{yx}) + \underbrace{O\left(\frac{1}{N}\right)}_{-\frac{1}{N}\operatorname{Tr}(UU).} \tag{33}$$

this is on the right in (31) for $U(N)$.



Uncertain reading. A straight bracket underlines the right-hand side of (33). Two vertical strokes appear above the term $-\frac{1}{N}\operatorname{Tr}(UU)$.

Original source details: GELFAND-P000904

It follows that

$$\beta\, \partial_\mu \frac{\delta\langle\Psi\rangle}{\delta\sigma_{\mu\nu}} = \oint dy_\nu\, \delta^4(x-y) \langle \Psi(c_{xy}) \Psi(c_{yx}) \rangle \tag{34}$$

We do not obtain closed equations.

c_{xy}, c_{yx} are closed because of $\delta^4(x-y)$

In fact, this is the first eq. of a chain of multiloop eqs. (one loop is insufficient) for Green functions.



Uncertain reading. The crossed-out entry before ‘We need’ may begin with «Не» (‘not’).

Original source details: GELFAND-P000904

We need to consider self-intersections of the contour as well

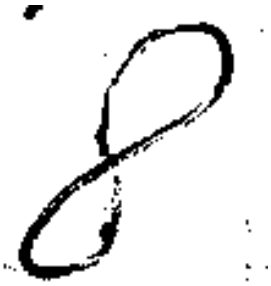
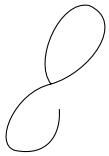


Diagram and typed description.
Self-intersecting contour.



Original source details: GELFAND-P000904

(they contain all the information about the interaction)

Trying to regularize the δ-function to give the equation meaning makes it meaningless (the contours open) — this must be done simultaneously with closing the contours.

$$\langle \Psi(c_1)\Psi(c_2)\dots\Psi(c_n)\rangle = \langle\Psi(c_1)\rangle\langle\Psi(c_2)\rangle\dots\langle\Psi(c_n)\rangle + O(N^{-2}) \tag{35}$$



Uncertain reading. The final argument on the right of (35) is overwritten. c_n is the preferred reading.

Original source details: GELFAND-P000904

$\Psi = \mathrm{Tr}\, U(c); \quad \Psi(0) = N.$

We must redefine: $\Psi = N\bar{\Phi}$.

For $\bar{\Phi}$ we obtain a certain eq., from which we get confinement



Uncertain reading. The crossed-out word below the last line is illegible. Two dots are visible to its right.

Original source details: GELFAND-P000904



Uncertain reading. The separate heavily crossed-out entry at the bottom is illegible.

Original source details: GELFAND-P000904

GELFAND-P000905

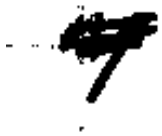
28.1.80. Gelfand Sem.

A. Zelevinsky. On the work of Kazhdan–Lusztig, reprs. of Coxeter groups, etc.

D. Kazhdan, G. Lusztig, Inv. Math, 53, $N = 2$, 1979

W — Coxeter group, S — simple reflections

W is generated by the generators S (a finite set).



Uncertain reading. A mark is crossed out between ‘finite’ and ‘set’. Its reading is unresolved.

Original source details: GELFAND-P000905

$$W = \langle S \mid s^2 = e,\ (ss')^{m(s,s')} = e \rangle \tag{1}$$

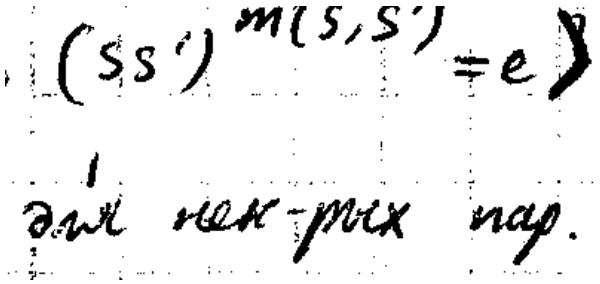
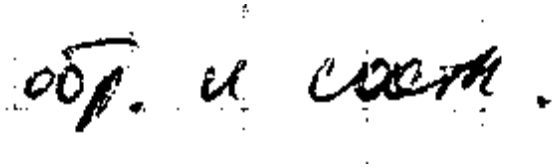


Diagram and typed description. Annotation beneath the second relation in (1).

/
для нек-рых пар.

Original source details: GELFAND-P000905

A Coxeter group is a group with such gens. and rels.



Uncertain reading. The last abbreviation reads «соотн.» (relations).

Original source details: GELFAND-P000905

Examples:

1) S_n: $\quad S = \{(i, i+1)\}$

$$((i, i+1)(i+1, i+2))^3 = e.$$

all the rest follow from these.

2) Weyl group W of a system of simple roots

$$w = s_1 \dots s_\ell \qquad \text{reduced, if the length is minimal} \qquad (2)$$
$$\ell = \ell(w)$$

If we discard any subset of the factors in a reduced expr. and multiply the rest, we obtain (def) the set of elements $\leq w$

$$w' = s_1 \dots \hat{s}_{i_1} \dots \hat{s}_{i_n} \dots s_\ell < w \qquad (3)$$



Uncertain reading. The last index on a deleted factor in (3) resembles i_n. The second letter is indistinct.

Original source details: GELFAND-P000905

This is independent of the reduced expression.

$$\mathcal{A} = \mathbb{Z}[q^{1/2}, q^{-1/2}]$$

Hecke algebra:

$\mathcal{H} \;\|\; \mathcal{H}_W$ — algebra over $\mathcal{A}$ with generators T_s, $s \in S$ and relations:

$$(T_s + 1)(T_s - q) = 0, \quad \forall s \qquad (4)_1$$

$$\underbrace{T_s T_{s'} T_s T_{s'} \dots}_{m(s,s')} = \underbrace{T_{s'} T_s T_{s'} T_s \dots}_{m(s,s')} \qquad (4)_2$$



Uncertain reading. A faint sloping stroke to the right of $(4)_2$ resembles part of a closing bracket.

Original source details: GELFAND-P000905

$q = 1 \Rightarrow$ we obtain the group algebra of the Coxeter group, since rels. (1) are written in the form

$$\underbrace{ss's's' \dots}_{m(s,s')} = \underbrace{s's \dots}_{m(s,s')} \qquad (5)$$



Uncertain reading. The crossed-out mark after the right-hand product in (5) is illegible.

Original source details: GELFAND-P000905

q — number of elements of a finite field; G — Chevalley grp. over q,

B — Borel subgroup

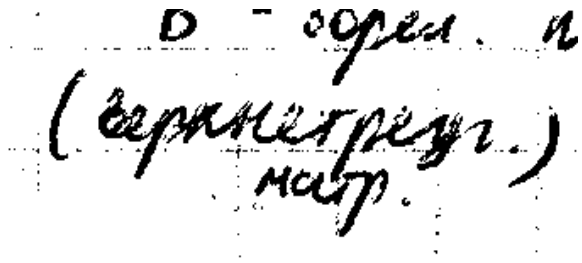


Diagram and typed description. Two-line annotation below ‘Borel subgroup’.

(upper triangular)
matrices

Original source details: GELFAND-P000905

$$\Rightarrow \quad \mathcal{H}(G,B) \cong \mathcal{H} \otimes C.$$

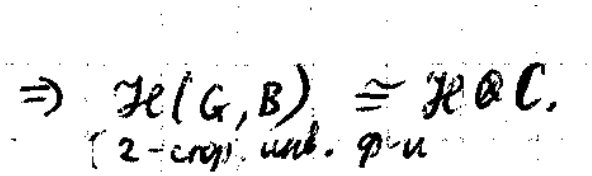


Diagram and typed description. Annotation under $\mathcal{H}(G,B)$.

(bi-inv. functions

Original source details: GELFAND-P000905

GELFAND-P000906

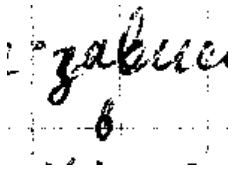
[source number: 8]

A basis is constructed in the alg. $\mathcal{H}$ (over $\mathcal{A}$)

Natural basis:

$$w = s_1 \dots s_\ell \rightsquigarrow \quad T_w = T_{s_1} \dots T_{s_\ell} \tag{6}$$

reduced (independent of the repr.)



Uncertain reading. A separate mark resembling Cyrillic «в» is visible under ‘independent’.

Original source details: GELFAND-P000906

$\{T_w\}$ — a basis in $\mathcal{H}$ (well known)



Uncertain reading. The crossed-out entry above ‘define’ may begin with «По».

Original source details: GELFAND-P000906

$$\text{Define an involution } \bar{\ } \text{ in } \mathcal{H}: \quad \overline{q^{1/2}} = q^{-1/2}, \quad \bar{T}_w = T_{w^{-1}}^{-1} \quad \left(\underset{\text{def}}{=} (T_{w^{-1}})^{-1}\right). \tag{7}$$

Invertibility follows from eq. $(4)_1$:

$$T_s^2 + T_s(1-q) = q, \quad \text{whence}$$

$$T_s^{-1} = q^{-1}(T_s + 1 - q) \tag{8}$$

Set

$$\varepsilon_w = (-1)^{\ell(w)}, \qquad q_w = q^{\ell(w)} \tag{9}$$

Theorem 1.1. $\forall w \in W,\ \exists!\ C'_w \in \mathcal{H}$:

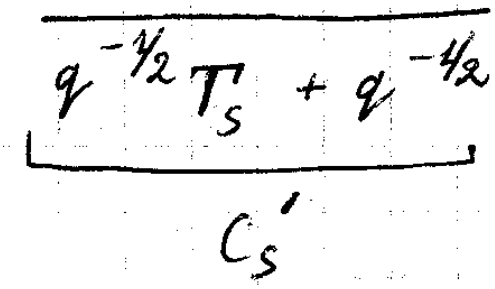
$$(1) \quad \overline{C'_w} = C'_w \tag{10}$$

$$(2) \quad C'_w = q_w^{-1/2} \sum_{y \le w} P_{y,w} \cdot T_y, \tag{11}$$

where

$$P_{w,w} = 1, \qquad \underset{(y<w)}{P_{y,w}} \text{ — polynomial in } q \text{ with coeffs. in } \mathbb{Z} \text{ of degree } \le \frac{1}{2}(\ell(w) - \ell(y) - 1) \tag{12}$$

$$\begin{aligned} \bar{T}_s &= q^{-1}T_s + q^{-1} - 1 \\ \overline{q^{-1/2}T_s} &= q^{-1/2}T_s + q^{-1/2} - q^{1/2} \\ \underbrace{\overline{q^{-1/2}T_s + q^{-1/2}}}_{C'_s} &= q^{-1/2}T_s + q^{-1/2} \end{aligned}$$



Original source details: GELFAND-P000906

Uncertain reading. In the third line, a straight bracket beneath the expression on the left is labelled C'_s.

Or: $P_{s,s} = 1, \quad P_{e,s} = 1.$

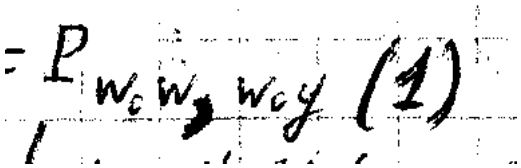


Original source details: GELFAND-P000906

Uncertain reading. The first index on the second polynomial is heavily traced over and is read as e.

Conjecture. Let g be a complex semisimple Lie algebra, W the Weyl group;

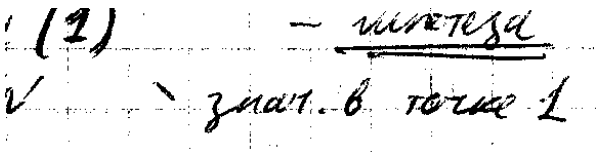
$$M_w \text{ — Verma module;} \qquad [M_w : L_y] = P_{w_0 w, w_0 y}(1) \quad \text{— conjecture} \tag{13}$$



Original source details: GELFAND-P000906

Uncertain reading. The indices read $w_0 w, w_0 y$. Both small zero glyphs have an open contour.

L_w — the unique irred. quotient, $\quad L_y,\ y \le w$



Original source details: GELFAND-P000906

Diagram and typed description. Annotation beside (1) in (13).

↘ value at 1

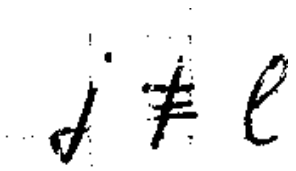
Checked for GL(4).

Vogan gave several equiv. reformulations.

$$P_{y,w} = \sum_{i \ge 0} q^i \dim \mathrm{Ext}^{\ell(w)-\ell(y)-2i}(M_y, L_w) \tag{14}$$

or

$$\mathrm{Ext}^j(M_y, L_w) = 0, \quad \text{if } j \not\equiv \ell(w) - \ell(y) \pmod 2 \tag{15}$$



Original source details: GELFAND-P000906

Uncertain reading. In (15) a diagonal stroke crosses three horizontal bars.

Set:

$$C_w = \varepsilon_w j(C'_w), \quad \text{where } j(q^{1/2}) = q^{-1/2}, \quad j(T_w) = \varepsilon_w q_w^{-1} T_w \tag{16}$$

(j — auxiliary involution)

$$y \prec w \overset{\text{def}}{\iff} y < w, \quad \varepsilon_y = -\varepsilon_w, \quad \deg P_{y,w} = \frac{1}{2}\big(\ell(w) - \ell(y) - 1\big). \tag{17}$$

Then denote by

$$\mu(y,w) \quad \text{the leading coefficient of } P_{y,w}.$$

Theorem 1.3. $s \in S,\ v \in W \Rightarrow$

$$T_s C_v = \begin{cases} -C_v, & sv < v, \\ qC_v + q^{1/2} C_{sv} + q^{1/2} \sum\limits_{\substack{z<v \\ sz<z}} \mu(z,v) C_z \end{cases} \tag{18}$$

if $sv > v$

(The action of the generators on the basis is written explicitly)

From this one can decompose the reg. repr. (block triangular form).

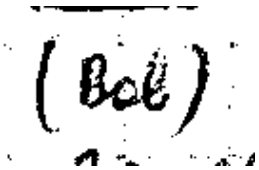
B. Mc Pherson (?) Макферсон.



Original source details: GELFAND-P000907

Uncertain reading. A sign, possibly an initial, is crossed out before «Макферсон».

(Bob)



Original source details: GELFAND-P000907

Uncertain reading. The annotation below the name is read tentatively as ‘Bob’.

Geom. interp. of $P_{s,t}(q)$

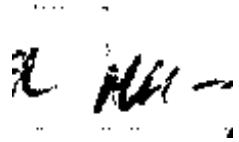
— q the Poincaré polyn. of strange cohomology of the Schubert cells.

The groups $IH_i(X)$ — intersection-homologie groups. (for sets with singularities)



Original source details: GELFAND-P000907

Uncertain reading. There is a crossed-out entry before ‘homologie’. Its reading is unresolved.



Original source details: GELFAND-P000907

Uncertain reading. A short entry is crossed out between «с» (with) and «особ.» (singularities).

(defined by the speaker)

their properties:

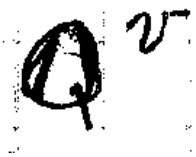
$$1) \quad X \text{ nonsing.} \quad \Rightarrow \quad IH_i(X) = H_i(X) \tag{1}$$



Uncertain reading. The abbreviation after «не» reads «синг.» (sing.).

Original source details: GELFAND-P000907

$$2) \quad IH_i(X) \times IH_{n-i}(X) \longrightarrow \mathbb{Q}^v \quad \text{when } \dim X = n. \tag{2}$$



Uncertain reading. Above the heavily traced letter $\mathbb{Q}$ is a mark resembling v.

Original source details: GELFAND-P000907

Simplest topological def.

$$X = X_n \supset X_{n-2} \supset X_{n-4} \supset \cdots \supset X_0 \tag{3}$$

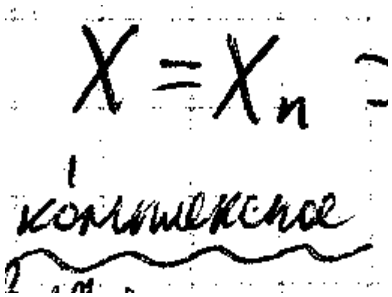


Diagram and typed description. Annotation below X in (3).

complex

Original source details: GELFAND-P000907

stratifications

Conditions:

a) $X_i - X_{i-2}$ — a nonsingular set of dim. i.

b) along X_{i-2} [inserted above: neighborhood] topologically triv. (locally) at points of $X_{i-2} - X_{i-4}$.



Diagram and typed description. Insertion above the line in condition b).

neighborhood

Original source details: GELFAND-P000907

X — Schubert cell (closed), dim. $n \Rightarrow$ the smaller cells can be used for stratification.
Then by def.

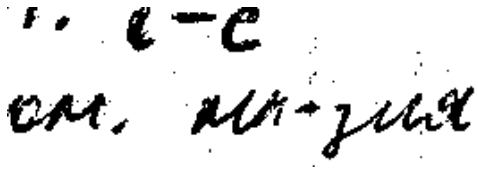


Original source details: GELFAND-P000907

Uncertain reading. A sign resembling I is crossed out above $IH_i(X)$.

$$IH_i(X) \underset{\text{def}}{=} \left\{ \begin{array}{c} i\text{-cycles meeting each } X_{n-2c} \\ \text{in a set of dimension } i-c-1 \\ \hline \partial \text{ of } i+1 \text{ chains meeting each } X_{n-2c} \\ \text{in a set of dim. } i-c \end{array} \right. \tag{4}$$

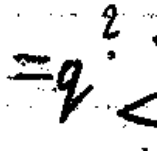
Theorem (using a special cohom. theory of sets of char. p)



Original source details: GELFAND-P000907

Uncertain reading. The abbreviation before ‘char.’ reads «мн-зий» (sets).

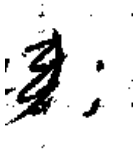
$$P_{s,t}(q) = q^{?} \sum_i \dim IH_i(X_t, X_t - y_s) q^i. \tag{5}$$



Original source details: GELFAND-P000907

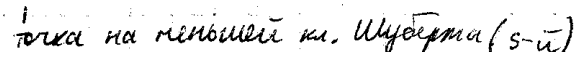
Uncertain reading. A question mark is written above the first q in (5).

s, t — elements of the Weyl group;



Original source details: GELFAND-P000907

Uncertain reading. A short mark is crossed out after ‘Weyl group’ and before the punctuation.



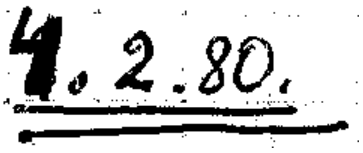
Original source details: GELFAND-P000907

Diagram and typed description. Annotation below y_s in (5).

|
point on the smaller Schubert cell (the sth)

GELFAND-P000908

4.2.80. Gelfand Sem.



Original source details: GELFAND-P000908

Uncertain reading. The first digit of the date is heavily traced over and reads as 4.

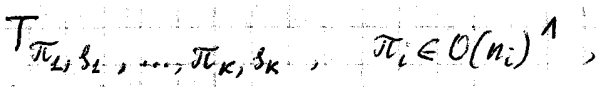
G. Olshanski.

$$G = \mathrm{GL}(\infty, \mathbb{R}) = \bigcup_{n \geq 1} \mathrm{GL}(n, \mathbb{R}) \tag{1}$$

$$K = \mathrm{SO}(\infty) = \bigcup_{n \geq 1} \mathrm{SO}(n) \tag{2}$$

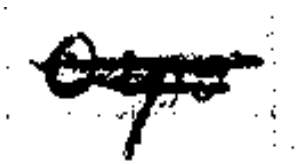
$\exists$ a family of reprs. of G for which K plays the role of a max. compact subgrp.

$$T_{\pi_1; s_1, \ldots, \pi_k, s_k}, \qquad \pi_i \in O(n_i)^\wedge, \quad s_i \in \mathbb{R}. \tag{3}$$



Original source details: GELFAND-P000908

Uncertain reading. Between π_1 and s_1 a comma and a separate dot above it are visible. The mark is rendered as a semicolon.



Original source details: GELFAND-P000908

Uncertain reading. The crossed-out abbreviation resembles «Огр» (restr.).

the restr. to K has discrete spectrum with finite multiplicities.

Analogous variant for $U(\infty)$.

$$\mathrm{SO}(p, \infty) = \overline{\bigcup_{n \geq 1} \mathrm{SO}(p, n)} \tag{4}$$

$$U(p, \infty) = \overline{\bigcup_{n \geq 1} U(p, n)} \tag{5}$$

$$\mathrm{Sp}(p, \infty) = \overline{\bigcup_{n \geq 1} \mathrm{Sp}(p, n)} \tag{6}$$

(completions)

(4)–(6) — type I groups, and a description of all irred. representations is given.



Original source details: GELFAND-P000908

Uncertain reading. Before ‘description’ the abbreviation reads «дан.» (given).

Mc Pherson.

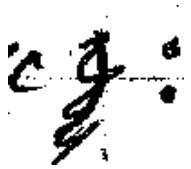
on the work of Gabrielov, Gelfand, and Losik on combin. Pontryagin classes,

which he understands better than the authors.

(on the first class)

X — a simplicial 4-manifold

(X — a simplicial complex g:



Original source details: GELFAND-P000908

Uncertain reading. After ‘complex’ there is a sign with a long descender, read tentatively as g.

$$\forall \text{ simplex } \sigma \quad \exists \text{ map} \quad \psi : \underset{\text{star of } \sigma}{\operatorname{St}\sigma} \longrightarrow \mathbb{R}^4, \tag{1}$$

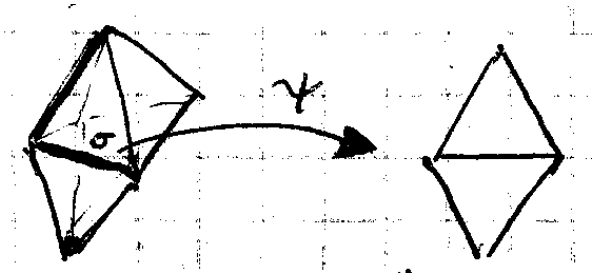


Original source details: GELFAND-P000908

Uncertain reading. A separate sign is crossed out before ψ.

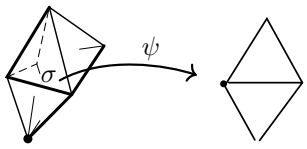
homeomorphic near the interior of σ, linear on each simplex.

The ideas are illustr. on a 2-manifold.



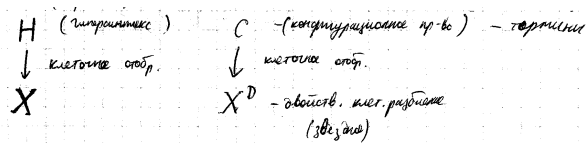
Original source details: GELFAND-P000908

Diagram and typed description. At left, a mesh with the front part σ marked and fine internal strokes. At right, two triangles with a common horizontal edge. The arrow is labelled ψ.



(def. of St)

H (hypersimplex) C — (configuration space) — terms.



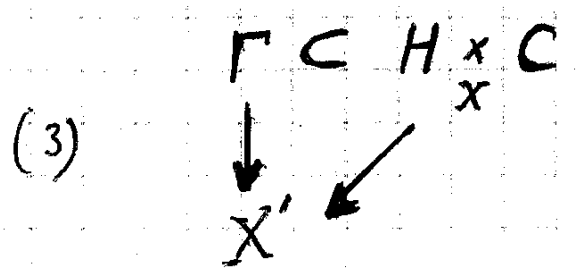
Original source details: GELFAND-P000908

Diagram and typed description. Cellular maps.

$$\begin{array}{ccc} & H & C \\ (2) & \big\downarrow \text{cellular map} & \big\downarrow \text{cellular map} \\ & X & X^D \end{array}$$

X^D — dual cell decomposition (star decomposition).

Cellularity: skeleton $\longrightarrow$ skeleton.



Original source details: GELFAND-P000908

Diagram and typed description. Inclusion into the fiber product and two maps to X'.

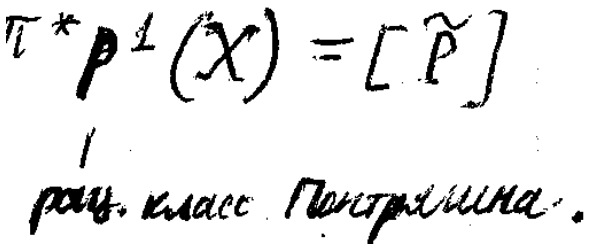
$$\begin{array}{cccc} (3) & \Gamma & \subset & H \underset{X}{\times} C \\ & \downarrow & \swarrow & \\ & X' & & \end{array}$$

GELFAND-P000909

[source number: 13]

$\widetilde{P}$ will be constr. — a cocycle in $C^4(\Gamma)$.

$$\pi^* p^1(X) = [\widetilde{P}]. \tag{4}$$



Original source details: GELFAND-P000909

Diagram and typed description. Annotation to $p^1(X)$ in (4).

rat. Pontryagin class.

Construction.

H

$\forall$ simplex σ in X, construct the polytope $H\sigma$ from maps

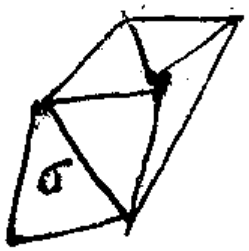
$$\{\text{vertices of the star of } \sigma\} \xrightarrow{h} [0,1]. \tag{5}$$

$$1) \quad h(v) = 1 \quad \text{when } v \in \sigma \tag{6}_1$$

$$2) \quad \sum_v h(v) = 5 = 4 + 1. \qquad (= \dim + 1) \tag{6}_2$$

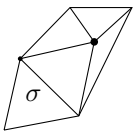
Examples.

a) simplex of highest dim. $\quad \operatorname{St}\sigma = \sigma$



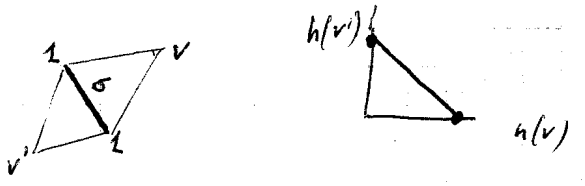
Original source details: GELFAND-P000909

Diagram and typed description. Star of a top-dimensional simplex, labelled σ.



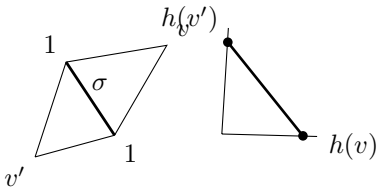
$H\sigma = 1$ point (the functions must be 1 everywhere).

b) simplex of codim. 1.



Original source details: GELFAND-P000909

Diagram and typed description. Two triangles with the common side σ, and a segment in the $h(v), h(v')$ plane.

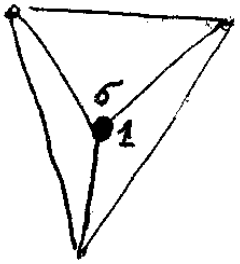


$H\sigma$ — a segment.

This also happens in the general case.

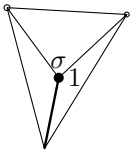
c) Codimension 2.

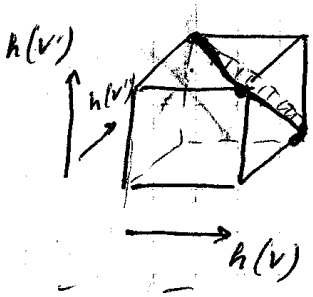
c′) (simple case)



Original source details: GELFAND-P000909

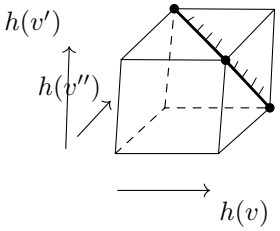
Diagram and typed description. Triangular mesh with the interior vertex marked σ and 1.





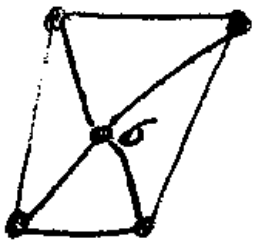
Original source details: GELFAND-P000909

Diagram and typed description. Cube section with three marked vertices and hatching. The arrows are labelled $h(v)$, $h(v')$, $h(v'')$.



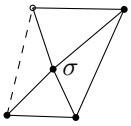
$H\sigma$ — a simplex. (2-dimensional simplex)

c″)



Original source details: GELFAND-P000909

Diagram and typed description. Quadrilateral mesh with σ at the interior vertex.



Cut the 4-dimensional cube by a plane equidist. from (some) vertices



Original source details: GELFAND-P000909

Uncertain reading. Below ‘cube’ a separate mark resembles 2.



Original source details: GELFAND-P000909

Uncertain reading. The crossed-out word before ‘equidist.’ is illegible. abc is written below it, beside a vertex of the drawing.



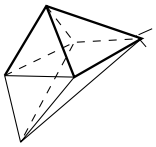
Original source details: GELFAND-P000909

Uncertain reading. The word after the cancellation is abbreviated ‘equidist.’.

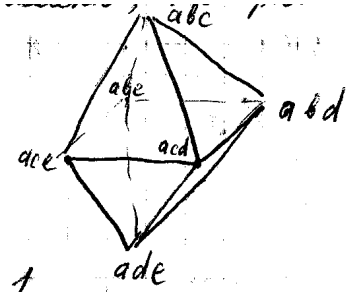


Original source details: GELFAND-P000909

Diagram and typed description. Octahedron with emphasized upper edges and broken interior strokes.

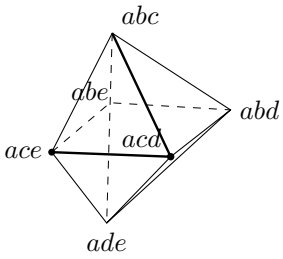


— an octahedron.



Original source details: GELFAND-P000909

Diagram and typed description. Octahedron with six labels, including abe by the broken rear connection.



Labels at the vertices of the polytopes.
The vertices are functions taking only the values 0 and 1.
The indices are those vertices where the value is 1.

$$H\sigma \times \sigma \qquad \sigma < \sigma' \quad \Rightarrow \quad H(\sigma') < H(\sigma) \quad \text{(the stars are nested this way)}$$

hence there is a common part

$$H\sigma \times \sigma \quad \text{and} \quad H\sigma' \times \sigma', \quad \text{namely}$$

$$\underline{H\sigma' \times \sigma}$$

Glue along this common part.
This gives the space H.
[Crossed out: Let us consider.]



Original source details: GELFAND-P000909

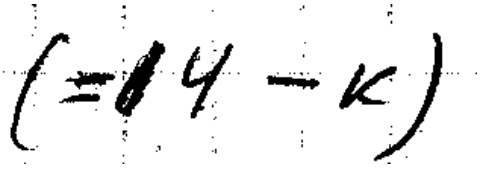
Uncertain reading. The crossed-out word is read tentatively as ‘Let us consider’.

GELFAND-P000910

Construction of C.
First we construct $C\sigma$.
By def. there is a map (at least one)

$$\operatorname{st}\sigma \xrightarrow{\psi} \mathbb{R}^4, \tag{7}$$

satisfying the conditions listed above.
$C\sigma$ — the set of all such ψ
Let σ have codimension k. $(= 4 - k)$



Original source details: GELFAND-P000910

Uncertain reading. In the parentheses after the equals sign and before $4 - k$, a short stroke resembling 1 is crossed out or traced over.

ψ is in general pos. if all sets

$$\left\{ \underbrace{v_1 \dots v_i}_{\text{all vertices of } \sigma} \qquad \underbrace{v'_1 \dots v'_k}_{k \text{ vertices from st}\,\sigma \setminus \sigma} \right\} \tag{8}$$

(5 points in all)

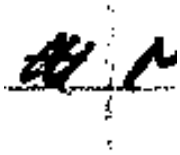
are in general position in $\mathbb{R}^4$.

ψ has speciality c if $\exists\, c$ sets for which this condition fails.



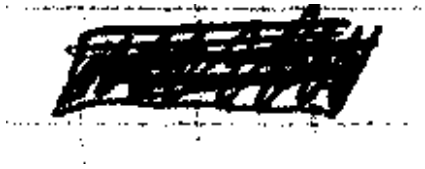
Original source details: GELFAND-P000910

Uncertain reading. The annotation beneath the second c is read tentatively as «набор» (collection). Its end is heavily traced over.



Original source details: GELFAND-P000910

Uncertain reading. An entry, possibly «и» (and), is crossed out after the second c.

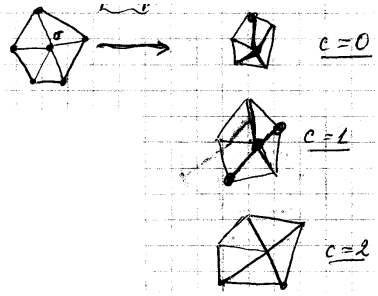


Original source details: GELFAND-P000910

Uncertain reading. The crossed-out word before ‘condition’ is illegible.

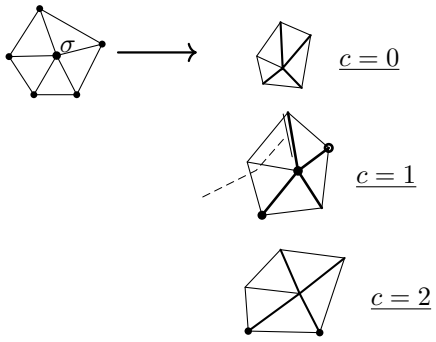
It is unknown: a stratification or a cell decomposition.

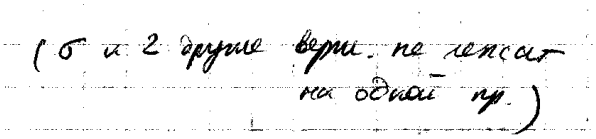
Example:



Original source details: GELFAND-P000910

Diagram and typed description. Five vertices around σ and three configurations. The faint sloping stroke at left in case $c = 1$ is drawn dashed.





Original source details: GELFAND-P000910

Diagram and typed description. Annotation beside $c = 0$.
(σ and 2 other vertices do not lie on one line)

$C\sigma$ for points $(k = 4)$

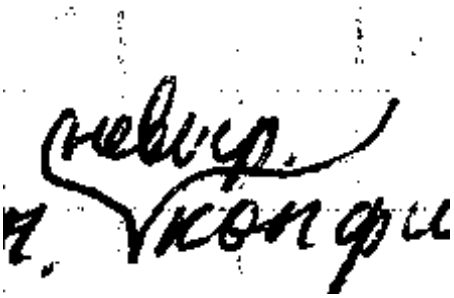
$C\sigma$ — 1 point for each connected comp. of configurations in general position

(discrete space of points)

$C\sigma$ for edges $(k = 3)$

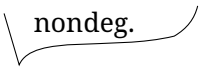
$C\sigma$ will be a graph

$C\sigma \ni$ a point for $\forall$ conn. comp. of [inserted above: nondeg.] configurations.



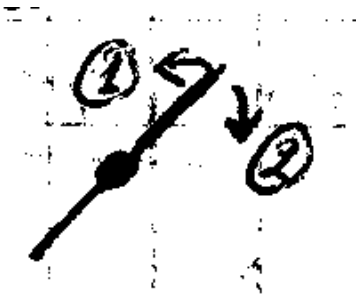
Original source details: GELFAND-P000910

Diagram and typed description. Insertion before ‘configurations’.



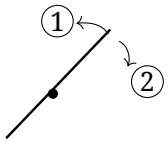
1 edge for each conn. component of the space of configurations with speciality 1

$c = 1 \Rightarrow \exists$ exactly 2 ways of bringing them into general pos.



Original source details: GELFAND-P000910

Diagram and typed description. Two arrows from the marked configuration.



There are no loops



Original source details: GELFAND-P000910

Diagram and typed description. A loop in parentheses.



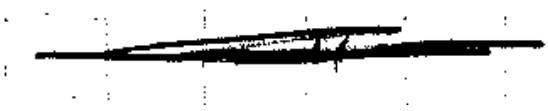
GELFAND-P000911

[source number: 14]

The graph is connected (a difficult theorem!)

$k = 2 \Rightarrow$ 2-dimensional (non-simpl.) complex.

points —



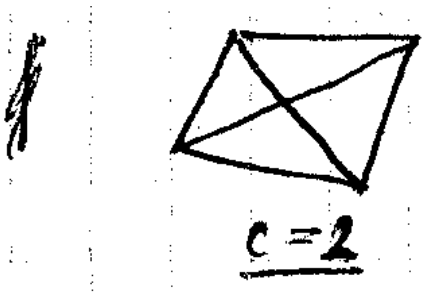
Original source details: GELFAND-P000911

Uncertain reading. Traces of writing appear under the long stroke. It may be a cancellation.

for connected comp. of config. in general pos.

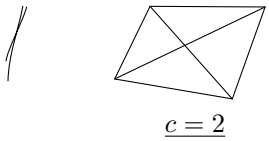
edges for config. of speciality 1

square for connected comp. —〃— 2



Original source details: GELFAND-P000911

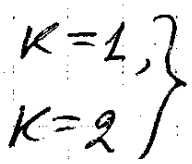
Diagram and typed description. Quadrilateral with both diagonals and the underlined label $c = 2$. The purpose of the double slanted stroke at left is uncertain.



break each one in 2 ways

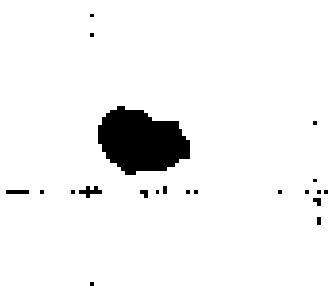
— these are the sides of the square.

$$\left.\begin{matrix} k=1, \\ k=2 \end{matrix}\right\} C\sigma \underset{\text{def}}{=} 1 \text{ point.}$$



Original source details: GELFAND-P000911

Uncertain reading. The second line reads $k = 2$.

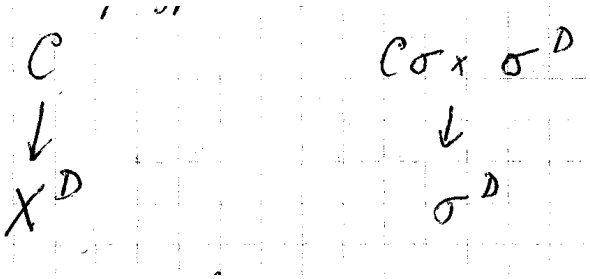


Original source details: GELFAND-P000911

Diagram and typed description. Isolated dot to the right of the definition.

•

(all config. in general pos. and form a connected set)

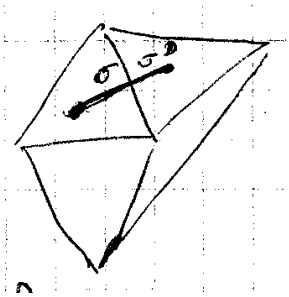


Original source details: GELFAND-P000911

Diagram and typed description. Two maps.

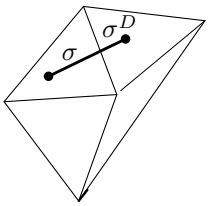
$$\begin{array}{ccc} C & & C\sigma \times \sigma^D \\ \downarrow & & \downarrow \\ X^D & & \sigma^D \end{array}$$

[At right:] do they glue together naturally?



Original source details: GELFAND-P000911

Diagram and typed description. Thickened segment with two marked ends and labels σ and σ^D. The faint interior edge on the right has a slight gap at the interior vertex.

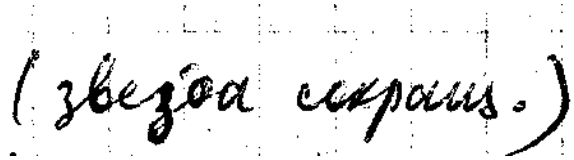


$$\sigma^D \subset \sigma'^D$$
$$\sigma \supset \sigma'$$

Then there is a map

$$\text{config. for } \sigma \longleftarrow \text{config. for } \sigma'$$

[At right:] (star shrinks)



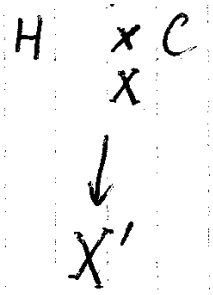
Original source details: GELFAND-P000911

Uncertain reading. The abbreviated Russian word is tentatively read «сокращ.» (shrinks).

(restriction of the maps ψ)

$$C\sigma \longleftarrow C\sigma'$$

We have a map:



Original source details: GELFAND-P000911

Diagram and typed description. Map to X'.

$$\begin{array}{c} H \underset{X}{\times} C \\ \downarrow \\ X' \end{array}$$

[At right:] $X' = X$ is decomposed into intersections of simplices and of the dual decomp.

$$\Gamma \subset H \underset{X}{\times} C$$

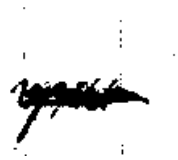
Γ — cells from H (a face of the hypersimplex)

for a vertex $\rightsquigarrow$ a point of the config. in general pos. $\Rightarrow$ then we include the direct products in Γ

σ — a simplex from X

$H\sigma$ $\quad$ $C\sigma$

face of $H\sigma \backsim$ [crossed out: illegible] set of five points $\xrightarrow{\psi}$ a set of five. If in general pos., include it in Γ



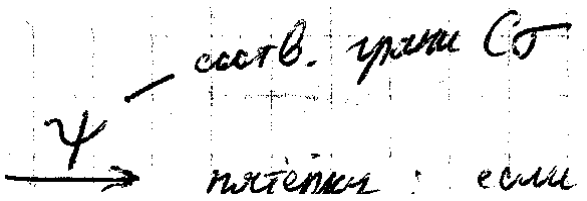
Original source details: GELFAND-P000911

Undeciphered passage. Short crossed-out word before ‘set of five’. It may be ‘face’.



Original source details: GELFAND-P000911

Uncertain reading. Two sloping strokes and a separate dot at the left of the upper stroke. The meaning of the sign is uncertain.

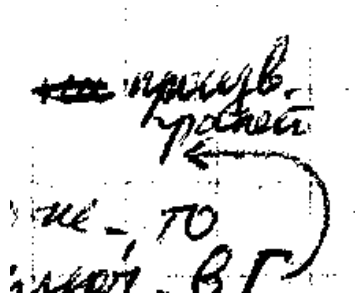


Original source details: GELFAND-P000911

Diagram and typed description. Sloping annotation line from ψ.

corresp. face of $C\sigma$

$\xrightarrow{\psi}$



Original source details: GELFAND-P000911

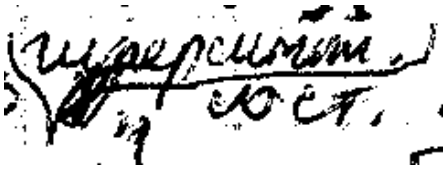
Diagram and typed description. Right-margin annotation. The short crossed-out mark before ‘prod.’ is illegible. It may be «c». The curved arrow comes from the lower line about Γ.

[crossed out] prod.

direct

$$\pm\frac{1}{48}\widetilde{P} \qquad \text{over the points of } X$$

Consider the 4-dimensional face [inserted above: hypersimpl.] consisting of the following maps

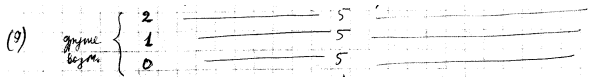


Original source details: GELFAND-P000912

Uncertain reading. There is a heavy stroke below the start of the insertion. It is unclear whether it cancels a separate entry.

$$h : \{\text{vert. of the star of } \sigma\} \longrightarrow [0,1]$$
$$h = 1 \text{ on } \sigma$$

$h = 1$ on 3 vertices (other ones), variable on 5, $= 0$ on the rest



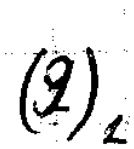
Original source details: GELFAND-P000912

Diagram and typed description. Three rows under the brace in (9).

$$(9)\ \text{other poss.}\ \begin{cases} 2 \text{ ____ } 5 \text{ ____} \\ 1 \text{ ____ } 5 \text{ ____} \\ 0 \text{ ____ } 5 \text{ ____} \end{cases}$$

(there are no other 4-dimensional faces)

$$\widetilde{P} = 0 \quad \text{. on } (9)_1 \text{ and } (9)_4$$
$$\widetilde{P} = \pm\frac{1}{48} \quad \text{on } (9)_2 \text{ and } (9)_3.$$

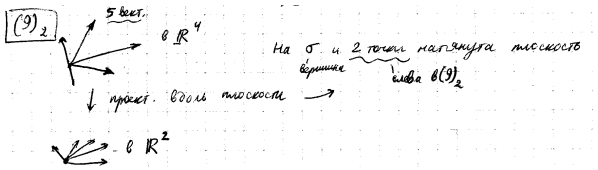


Original source details: GELFAND-P000912

Uncertain reading. The first label is tentatively $(9)_1$. There is an extra horizontal stroke below the 9 inside the parentheses.

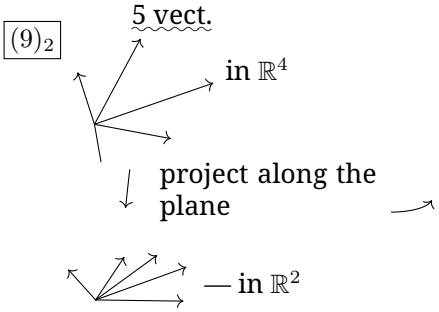
The number refers to an oriented cell.

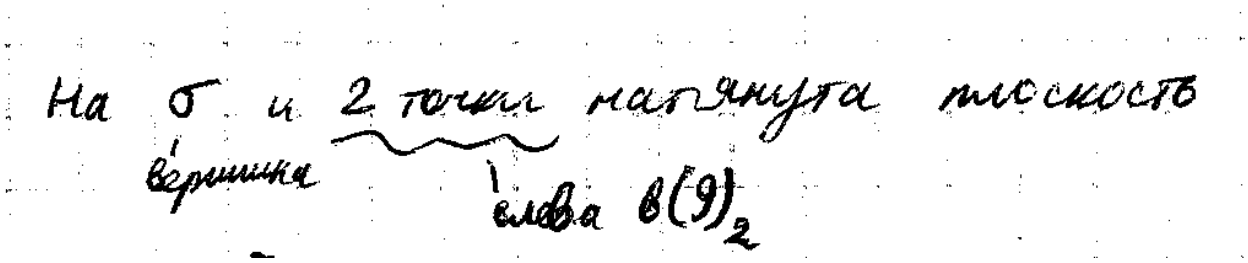
To the orientation of face $(9)_2$ corresponds the parity of the ordering of the 5 points.



Original source details: GELFAND-P000912

Diagram and typed description. Five vectors and their projection. The lowest stroke in the top group has no visible arrowhead.

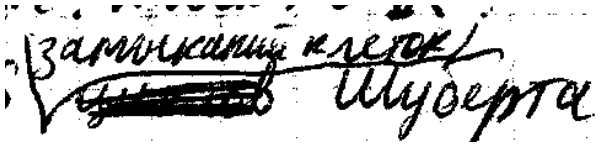




Original source details: GELFAND-P000912

Diagram and typed description. Annotation at the right of the figure.
The plane is spanned by σ [label below: vertex] and 2 points [label below: at left in $(9)_2$].

10 orientations of the plane (each pair)
We must take $(-1)^i \frac{1}{48}$, where i is the number of positive orientations
(it does not matter which are > 0, since there are 10 in all)
$\boxed{(9)_3}$ the same thing
we get 5 vectors in $\mathbb{R}^3$,
also 10 orient. in $\mathbb{R}^3$, and so on.
$G_{2,2}(\mathbb{R})$ — 2-dimensional planes in $\mathbb{R}^4$
there are 6 [crossed out: cycles?] [inserted above: closures of cells] of Schubert



Uncertain reading. A word beneath the insertion is crossed out, tentatively «циклов» (cycles).

Original source details: GELFAND-P000912

In $\mathbb{R}^4$: [mark crossed out] 6 coordinate 2-dimensional planes



Uncertain reading. The mark before 6 is heavily crossed out. The reading is unresolved.

Original source details: GELFAND-P000912

Planes [entry crossed out; inserted: not] transverse to the given coordinate plane



Uncertain reading. The crossed-out entry before the insertion ‘not’ is illegible.

Original source details: GELFAND-P000912

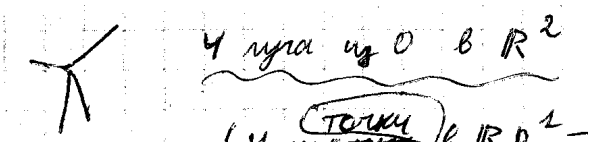
“enhanced projective configur. of 4 points in $\mathbb{R}P^2$”



Uncertain reading. The exponent in the English phrase reads 2.

Original source details: GELFAND-P000912

“improved — improved [Russian gloss: «улучшенные»]

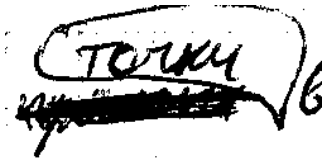


Original source details: GELFAND-P000912

Diagram and typed description. 4 rays from 0 in $\mathbb{R}^2$

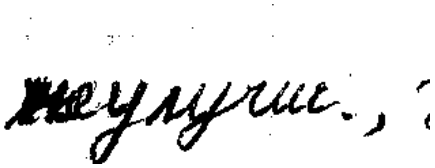


(4 [crossed out: illegible] [inserted above: points] in $\mathbb{R}P^1$ — unimproved, here rays)



Original source details: GELFAND-P000912

Undeciphered passage. The crossed-out word before the insertion ‘points’ is illegible.



Original source details: GELFAND-P000912

Uncertain reading. The beginning of «неулучш.» (unimproved) is heavily traced over.

GELFAND-P000913

[source number: 15]



Original source details: GELFAND-P000913

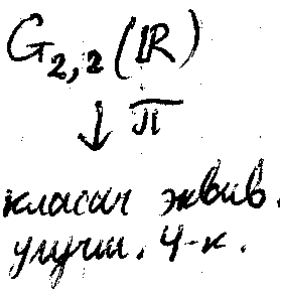
Diagram and typed description. Manuscript page number at upper right.

$$L \subset \mathbb{R}^4, \qquad \dim L = 2$$

in $\mathbb{R}^4/L$ there are 4 rays!!

We must take their cross ratio

2 such collections are equiv. if the action of $GL(2,\mathbb{R})$ carries one to the other



Original source details: GELFAND-P000913

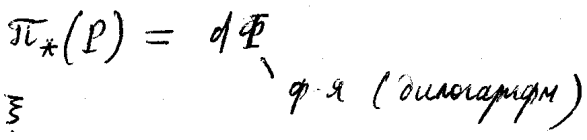
Diagram and typed description. Arrow labelled π.

$G_{2,2}(\mathbb{R})$

$\downarrow \pi$

classes of equiv.
enhanced quadruples

P — 4-form representing the Pontryagin class



Original source details: GELFAND-P000913

Diagram and typed description. Sloping annotation line from Φ.

$\pi_*(P) = d\Phi$

fn. (dilogarithm)

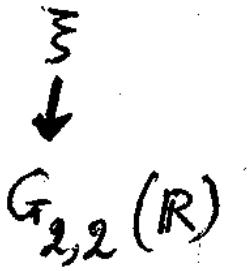


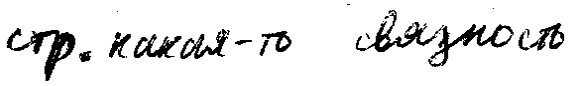
Diagram and typed description. Arrow from ξ to the Grassmannian.

$$\begin{array}{c} \xi \\ \downarrow \\ G_{2,2}(\mathbb{R}) \end{array}$$

Original source details: GELFAND-P000913

$P = p(\xi)$, ξ — 2-dimensional (std.) bundle over $G_{2,2}(\mathbb{R})$

construct some connection.



Uncertain reading. The initial Russian verb is abbreviated «стр.», read as ‘construct’.

Original source details: GELFAND-P000913

GELFAND-P000914

11.2.80 Gelfand Sem.

R. Mc Pherson.

The Grassmannian $G_{n,k}$ of n-planes in $\mathbb{R}^{n+k}$,

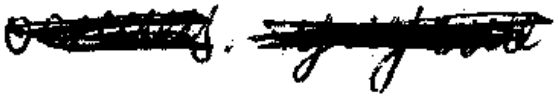
(unoriented)

Idea:

To an element of $G_{n,k}$ corresp. the pair

hypersimplex + configuration

$\widetilde{C}_{n,k}$ = [crossed out: enhanced?, [illegible]] enhanced project. configurations



Undeciphered passage. The first of the two crossed-out words may be «оснащ.» (enhanced). Two descending loops survive in the second word, which cannot be read in full.

Original source details: GELFAND-P000914

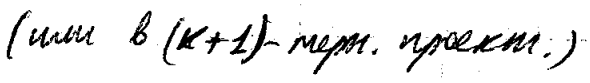


Diagram and typed description. The English word ‘enhanced’ is written beneath the Russian abbreviation «оснащ.».

оснащ.
enhanced

Original source details: GELFAND-P000914

of $n + k$ points in a k-dimensional space (or in a $(k + 1)$-dimensional project. one)



Uncertain reading. The parentheses clearly contain $k + 1$.

Original source details: GELFAND-P000914



Original source details: GELFAND-P000914

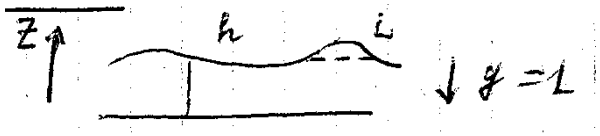
Undeciphered passage. Only isolated upper strokes remain below the last full line. No continuous text can be read there.

GELFAND-P000915

18.2.80 Gelfand Sem.

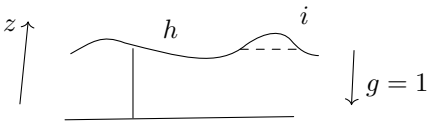
Zakharov V. E.

Benney eqs.



Original source details: GELFAND-P000915

Diagram and typed description. Two rises in the surface above the bed. A mark with a separate upper dot above the right crest is tentatively read i.



h — the shape of the surface, $h = h(\vec{r}, t)$.

$\vec{r}$ — a vector in the plane of the bed



Original source details: GELFAND-P000915

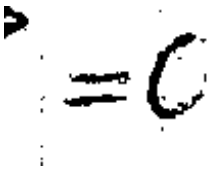
Uncertain reading. The horizontal coordinate reads $\vec{r}$.

$\vec{u} = \vec{u}(\vec{r}, z, t)$ — velocity in the plane of the bed

w — vert. velocity

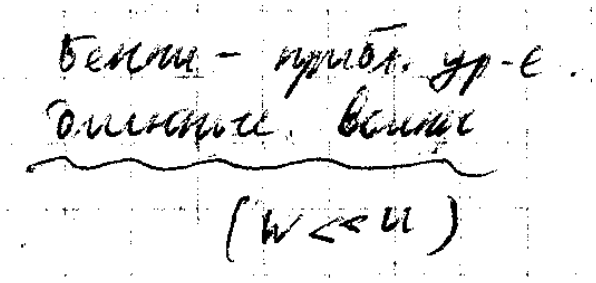
ideal incompress. fluid

$$\frac{\partial w}{\partial z} + \operatorname{div} \vec{u} = 0$$



Original source details: GELFAND-P000915

Uncertain reading. The glyph after the equals sign is open at the right. The context supports reading it as 0.



Original source details: GELFAND-P000915

Diagram and typed description. Annotation at the right margin.

Benney — approx. eq.

long waves

$(w \ll u)$

$$\frac{\partial w}{\partial t} + \underbrace{(u\nabla)w + w\frac{\partial w}{\partial z}}_{\text{neglect compared with the following terms}} - \frac{\partial p}{\partial z} + 1 = 0$$

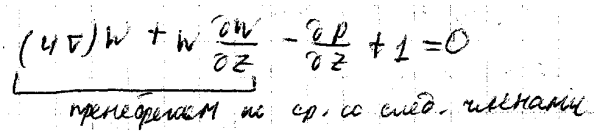
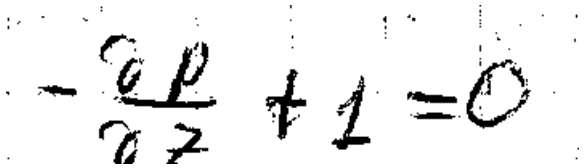


Diagram and typed description. A square bracket below the two transport terms.

Original source details: GELFAND-P000915



Uncertain reading. A minus sign is written before the pressure derivative.

Original source details: GELFAND-P000915

GELFAND-P000916

$$(1)\quad \begin{cases} \dfrac{\partial w}{\partial z} + \operatorname{div}\vec{u} = 0 \\ \vec{u}_t + (u\nabla)u + w\dfrac{\partial \vec{u}}{\partial z} + \nabla h = 0 \end{cases}$$

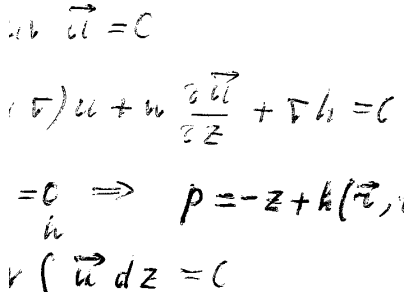
$$\frac{\partial p}{\partial z} + 1 = 0 \quad \Longrightarrow \quad p = -z + h(\vec{r}, t)$$



Uncertain reading. The coordinate in the argument of h reads $\vec{r}$.

Original source details: GELFAND-P000916

$$(1)_3 \quad \left\{ h_t + \operatorname{div}\int_0^h \vec{u}\,dz = 0 \right.$$



Uncertain reading. The four right-hand signs read as 0. Several have contours that are open at right.

Original source details: GELFAND-P000916

(1) — system of Benney eqs.

Manin [written below: and others] — the 1-dimensional case.

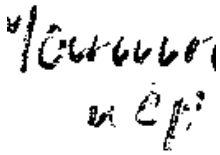


Diagram and typed description. ‘And others’ written below Manin’s name.

Original source details: GELFAND-P000916

Zakharov a) in the 1-dimensional case derived a countable number of conserv. laws, there are many more of them

Uncertain reading. The verb reads ‘derived’.

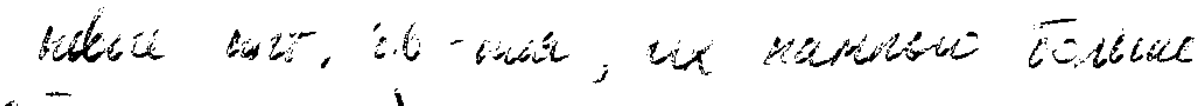


Original source details: GELFAND-P000916

Uncertain reading. The abbreviation after ‘countable’ is read «з.с-ний» (conservation laws).



Original source details: GELFAND-P000916



Original source details: GELFAND-P000916

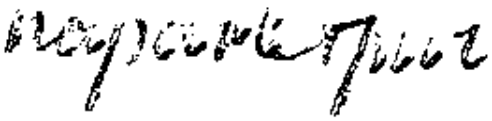
Diagram and typed description. End of item a). The final words read ‘there are many more of them’.

b) a different Ham. structure (simpler)

c) something for 2-dimensional dep. (with Hamiltonian character)

Idea: reinterpret. of the problem.

Uncertain reading. The abbreviation before ‘of the problem’ reads ‘reinterpret.’.



Original source details: GELFAND-P000916

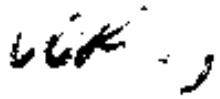
I.M.G. Collective diff. eqs.:

when the particles do not interact dir., but [crossed out: illegible] [inserted above: are charged and] together create a field which [between the lines: (electr. field)] acts on each one separately.



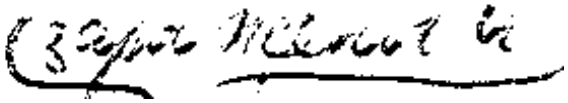
Undeciphered passage. A short entry after ‘but’ is crossed out. The reading is unresolved.

Original source details: GELFAND-P000916



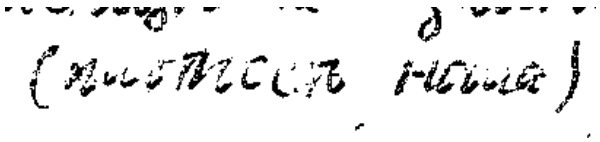
Original source details: GELFAND-P000916

Uncertain reading. The abbreviation after ‘interact’ is tentatively read «неп.» (dir.).



Original source details: GELFAND-P000916

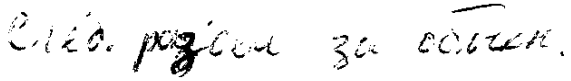
Diagram and typed description. The insertion ‘are charged and’ has a long leader.



Original source details: GELFAND-P000916

Uncertain reading. The first word in parentheses is tentatively read ‘electr.’.

Next section after ordinary diff. eqs.



Original source details: GELFAND-P000916

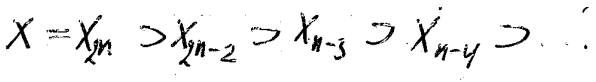
Uncertain reading. The reading ‘Next section after ordinary’ is tentative. There is a separate stroke above the last abbreviation.

GELFAND-P000917

22.9.80 Gelfand Sem.

Fuchs. There is a variety with sing. (algebr., complex) and there is a stratification.

$$(1) \quad X = X_{2n} \supset X_{2n-2} \supset X_{n-3} \supset X_{n-4} \supset \dots$$



Original source details: GELFAND-P000917

Uncertain reading. There is no 2 before n in the last two indices written out.

$X_i \setminus X_{i-1}$ — an i-dimensional variety.

Near a given stratum everything is arranged in the same way (the whole neighborhood).

Consider sing. simplices of the space X with certain additional properties.

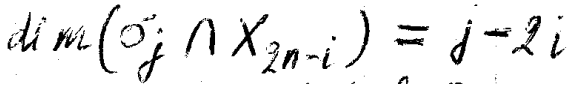
$\sigma_j \qquad \dim \sigma_j = j$



Original source details: GELFAND-P000917

Uncertain reading. An illegible mark between dim and σ_j is crossed out.

$$(2) \quad \dim(\sigma_j \cap X_{2n-i}) = j - 2i$$

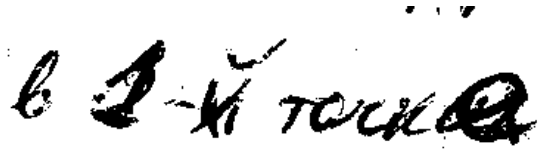


Original source details: GELFAND-P000917

Uncertain reading. In (2) the index on the left reads $2n - i$, and the expression on the right reads $j - 2i$.

There is Poincaré duality in these cohom. These are different cohomologies.

Example: Two 2-dimensional spheres glued at one point

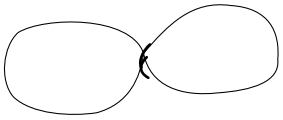


Uncertain reading. The digit reads 1. The end of the Russian word for ‘point’ is traced over.

Original source details: GELFAND-P000917



Diagram and typed description. Two outlines meeting at the center. The strokes are retraced in the narrow contact region.



Original source details: GELFAND-P000917

These cohomologies are not homotopy invariant.

Ordinary homology:

$$\begin{cases} 0 & \mathbb{Z} \\ 1 & 0 \\ 2 & \mathbb{Z} \dot{+} \mathbb{Z} \end{cases}$$



Uncertain reading. A short entry at right of the first row is crossed out. Its beginning resembles $\mathbb{Z}$, but the entire entry is illegible.

Original source details: GELFAND-P000917

They do not satisfy Poincaré duality (immediately apparent).

MacPherson hom.

$$\begin{cases} 0 & \mathbb{Z} \dot{+} \mathbb{Z} \\ 1 & 0 \\ 2 & \mathbb{Z} \dot{+} \mathbb{Z}. \end{cases}$$

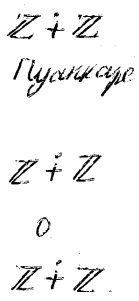


Diagram and typed description. In all three sums a dot is written above the plus sign.

Original source details: GELFAND-P000917

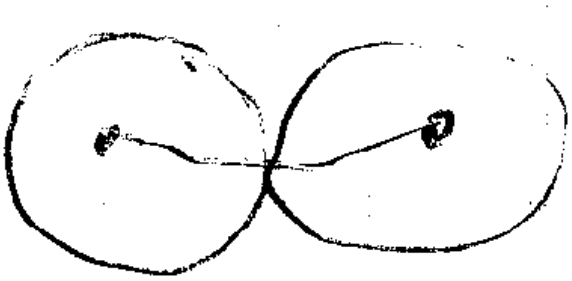
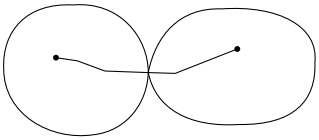


Diagram and typed description. A bent line joins two marked points and passes through the contact of the outlines.



Original source details: GELFAND-P000917

Such a simplex is forbidden.

Conjecture: they agree with L_2-cohomology.

(this interested Borel)

Suppose there is exactly one isol. singularity. Then MF-cohom.

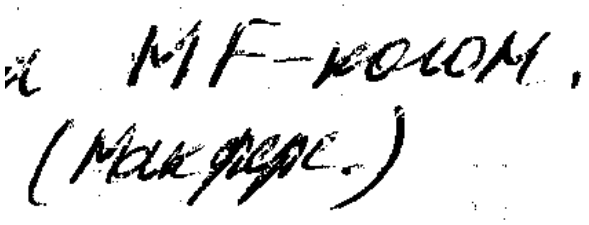


Diagram and typed description. '(MacPhers.)' is written under 'MF-cohom.'.

Original source details: GELFAND-P000917

$$\left.\begin{array}{ll} \text{in dim. } > \dfrac{\dim X}{2} & \text{agree with ordinary ones,} \\ \text{in dim. } < \dfrac{\dim X}{2} & \text{from Poincaré duality.} \end{array}\right] \quad \text{this is Deligne's def.}$$

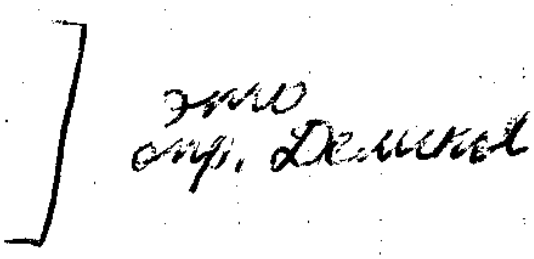


Diagram and typed description. 'This is Deligne's def.' is written to the right of the bracket.

Original source details: GELFAND-P000917

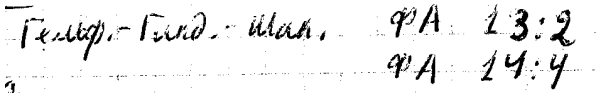
GELFAND-P000918

Deligne's def. in a more general situation (inductive)

M. I. Graev.

From the works of Gelf.-Gind.-Shap.

FA 13 : 2
FA 14 : 4

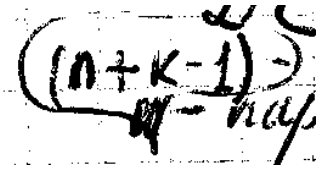


Uncertain reading. The last surname is abbreviated 'Shap.'. The right-hand entries read 'FA 13:2' and 'FA 14:4'.

Original source details: GELFAND-P000918

$\mathcal{D} \subset \mathbb{R}^n$,

$(n+k-1)$-param. family of curves



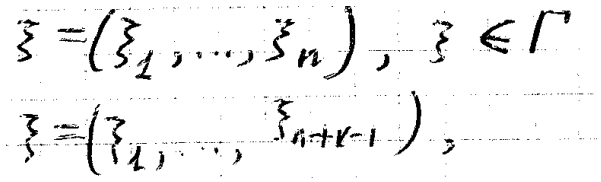
Original source details: GELFAND-P000918

Uncertain reading. $(n+k-1)$ is inserted above and linked to the crossed-out entry. The earlier mark is illegible.

$$x_i = \varphi_i(x_1, \xi), \qquad \xi = (\xi_1, \ldots, \xi_n), \qquad \xi \in \Gamma$$

Integr. transform

$$\xi = (\xi_1, \ldots, \xi_{n+k-1}),$$



Original source details: GELFAND-P000918

Uncertain reading. The first list ends at ξ_n, the second at ξ_{n+k-1}.

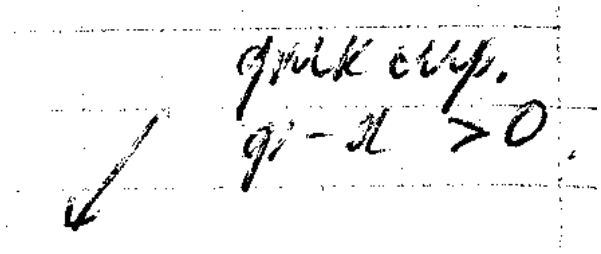
$f(x)$, supp $f \subset \mathcal{D}$

$$F(\xi) = I_\chi f = \int f(x_1, \varphi_1(x_1, \xi), \ldots, \varphi_n(x_1, \xi))\chi(x_1, \xi)\, dx_1$$



Original source details: GELFAND-P000918

Uncertain reading. The first index on φ under the integral reads 1.



Original source details: GELFAND-P000918

Diagram and typed description. An arrow from the annotation to the factor χ.

fixed everywh.
fn. > 0.

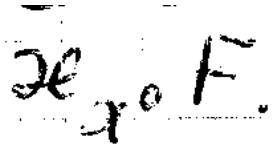
χ

It is of interest to find such sets of curves for which

$\exists$ a loc. inversion formula exists (recovery from integrals over curves passing through this point or an infinitesimally close one)

Such sets of curves are called admissible.

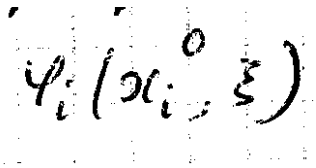
$\forall x^0 \in \mathcal{D}$ define an operator $F(\xi) \longrightarrow \mathcal{H}_{x^0} F$.



Original source details: GELFAND-P000918

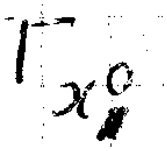
Uncertain reading. The operator's script letter is rendered $\mathcal{H}$.

$$\Gamma_{x^0} \qquad x_i^0 = \varphi_i(x_i^0, \xi) \qquad \dim \Gamma_{x^0} = k$$



Original source details: GELFAND-P000918

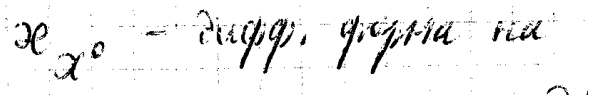
Uncertain reading. The dot on the subscript i is visible in the first argument of φ_i.



Original source details: GELFAND-P000918

Uncertain reading. An illegible mark below x in the subscript of the left-hand Γ is crossed out.

$\mathcal{H}_{x^0}$ — a diff. form on



Original source details: GELFAND-P000918

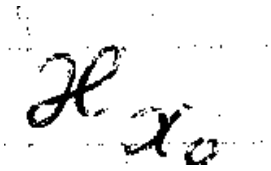
Undeciphered passage. The line ends with ‘on’. No object is written after it.

$$\mathcal{H}_{x^0} F = \sum a_{ij}(x_1^0, \xi) \frac{\partial F}{\partial \xi_j}\, d\xi_i + F \sum b_i(x_1^0, \xi)\, d\xi_i$$

The operator $\mathcal{H}_{x^0}$ is called admissible, if $\forall$ fn. f the diff. form
$\mathcal{H}_{x^0} I_\chi f$ is closed.
$k > 1 \Rightarrow$ this is the complete def.
If $k = 1$, it is automatically closed and it is req. that the functional

$$f \mapsto \int \mathcal{H}_{x_0} I_\chi f$$

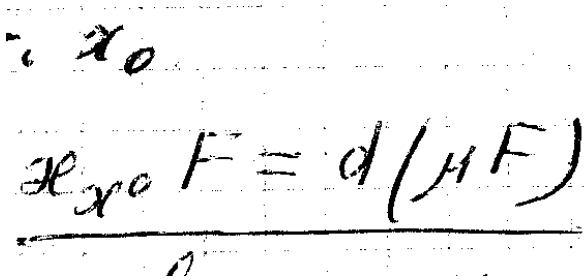
be concentrated at the point x_0.



Original source details: GELFAND-P000918

Uncertain reading. In the integral and the next line 0 is a subscript. In the preceding formulas the 0 on x is a superscript.

Trivial functional: $\mathcal{H}_{x^0} F = d(\mu F)$



Original source details: GELFAND-P000918

Diagram and typed description. The formula for the trivial functional is underlined.

The quotient by the triv. functionals is important.